Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions.

Made with GAMMA

Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

Review ratings imputed

Key features include customer demographics, purchase details, and shopping behavior metrics.

Made with GAMMA

Python Data Preparation

Data Loading & Exploration

Imported dataset using `pandas` and examined structure with `df.info()` and `.describe()`.

Missing Data Handling

Imputed 37 missing review ratings using median values by product category.

01

Column Standardization

Rename to snake case

02

Feature Engineering

Create age groups and purchase frequency

03

Database Integration

Load to PostgreSQL

Made with GAMMA

SQL Analysis: Revenue Insights

Revenue by Gender

Male customers generated $157,890 vs. female customers at $75,191

High-Spending Discount Users

839 customers used discounts while spending above average purchase amount

Top Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)

Made with GAMMA

Shipping & Subscription Analysis

Shipping Type Comparison

- Standard: $58.46 average
- Express: $60.48 average

Subscription Status

- Subscribers: 1,053 customers, $59.49 avg spend
- Non-subscribers: 2,847 customers, $59.87 avg spend

Made with GAMMA

Discount Strategy Insights

1

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%)

2

Customer Segmentation

Loyal: 3,116 customers, Returning: 701, New: 83

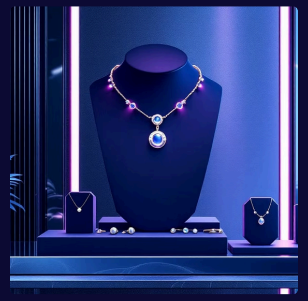
3

Repeat Buyers & Subscriptions

958 repeat buyers subscribed vs. 2,518 non-repeat buyers

Made with GAMMA

Top Products by Category



Accessories

Jewelry (171 orders), Sunglasses (161), Belt (161)

Clothing

Blouse (171 orders), Pants (171), Shirt (169)

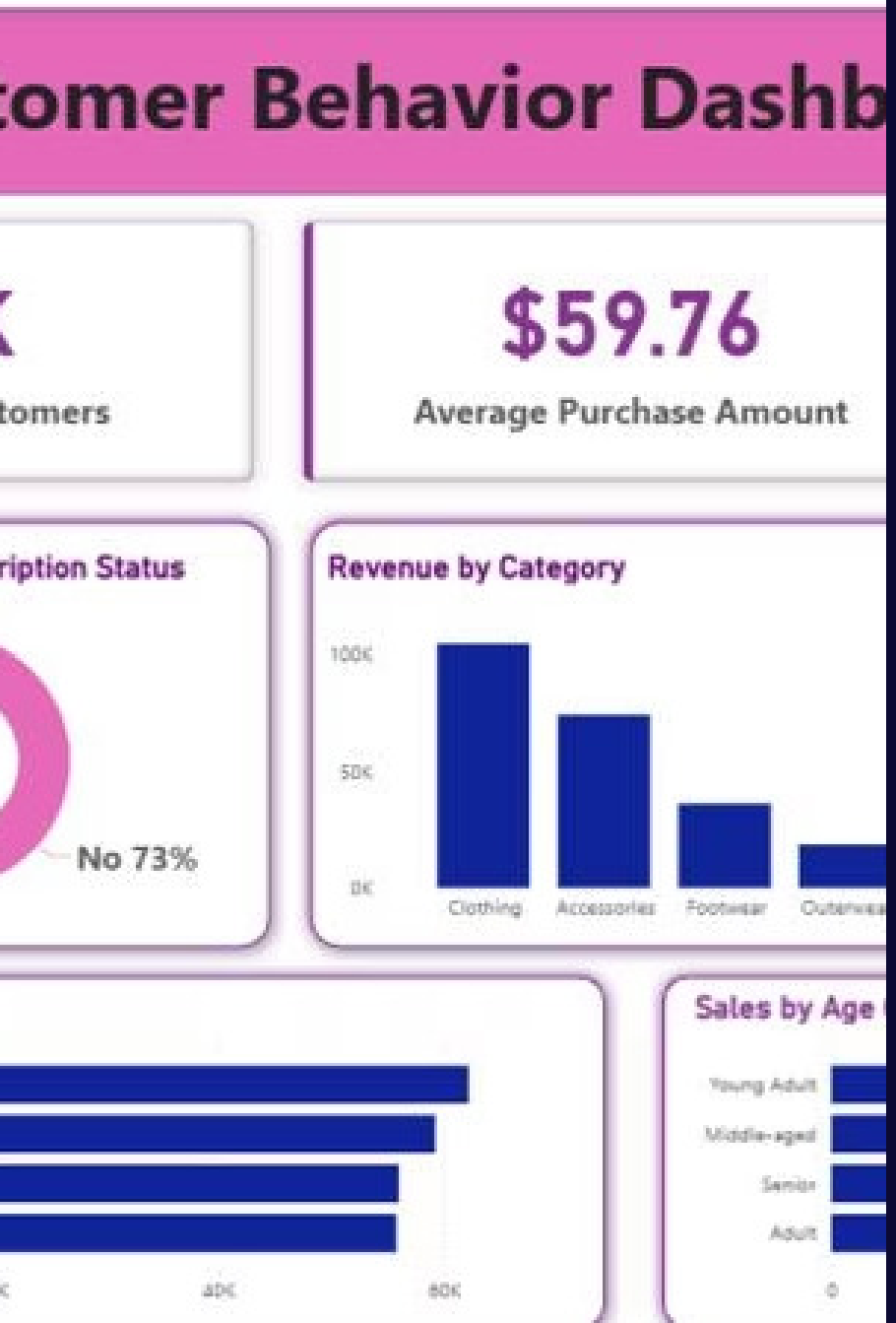
Footwear

Sandals (160 orders), Shoes (150), Sneakers (145)

Outerwear

Jacket (163 orders), Coat (161)

Made with GAMMA



Power BI Dashboard

KPIs

3,900 customers, $59.76 average purchase, 3.75 average rating

Filters

Subscription status, gender, category, shipping type

Visualizations

Donut charts, bar charts showing revenue and sales by category and age group

Made with GAMMA

Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

Made with GAMMA